CPEN 416 / EEC E 571O

Lecture 16: Expectation values, mixed states, and density matrices

Tuesday 4 November 2025

Announcements

kraus

$a^r \equiv 1 \bmod 31$

$26^6 \equiv 1$

- Quiz 7 today
- Practice problem set 4 partially available
- Friday office hours cancelled due to travel

$N = 31$

$a = 26$

$U_{31,26} |\underbrace{10010}_{18}\rangle$

$= |26 \cdot 18 \bmod 31\rangle$

$= |13 \bmod 31\rangle$

$= |01101\rangle$

Where are we going?

1. Basics of QC
2. Grover's algo (quad. speedup)
3. Shor's algo (exponential speedup)
4. Theory of QI (density matrices, channels)
5. Error correction

$|0\rangle_L \to |000\rangle$

$|1\rangle_L \to |111\rangle$

$X_L |0\rangle_L = |1\rangle_L$

$(X \otimes X \otimes X)|000\rangle$

Learning outcomes

- Define an observable and compute its expectation value
- Define *mixed states* and express quantum states as density matrices
- Identify mixed states on the Bloch sphere

Review: projective measurements

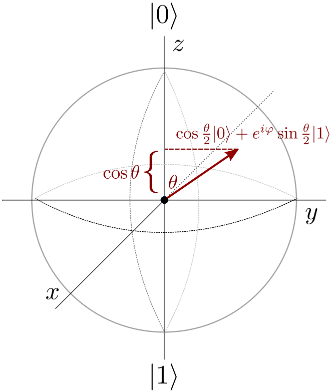
Recall: when we measure a qubit in state $|\varphi\rangle$ with respect to basis $\{|\psi_i\rangle\}$, the probability of observing it in state $|\psi_i\rangle$ (outcome i) is

$$\Pr(\text{outcome } i) = |\langle\psi_i|\varphi\rangle|^2$$

If we observe outcome i, the qubit *remains in* $|\psi_i\rangle$.

Review: projective measurements

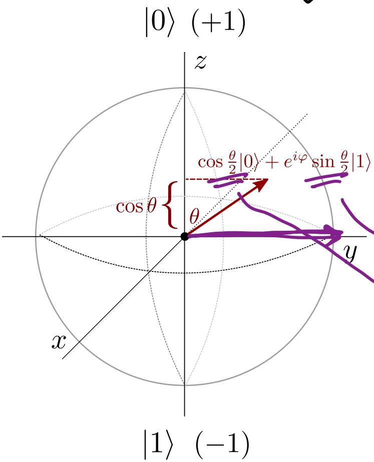
Projective measurement corresponds to a geometric projection onto the axis represented by a basis.



Let's look at measurement in a slightly different way...

Expectation values

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



Assign each basis states a real-valued outcome:

$$\boxed{|0\rangle \to +1 \qquad |1\rangle \to -1}$$

If we prepare and measure *N* times, what is the **expected value** of the meas. outcome?

$+1: N_+$

$-1: N_-$

$$1 \cdot \frac{N_+}{N} + (-1)\frac{N_-}{N}$$

In the limit of infinite shots...

$$1 \cdot p_0 + (-1) p_1$$
$$= \cos^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)$$
$$= \cos\theta$$

Expectation values

This assignment is no accident...

$|\alpha|^2 = \alpha\alpha^*$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

$|1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

Its eigensystem is

$\lambda_0 = +1 \qquad |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$\lambda_1 = -1 \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

exp. val. of Z

$$\begin{aligned} \langle Z \rangle &= \lambda_0 p_0 + \lambda_1 p_1 \\ &= \lambda_0 |\langle 0|\psi\rangle|^2 + \lambda_1 |\langle 1|\psi\rangle|^2 \\ &= \lambda_0 \langle 0|\psi\rangle\langle\psi|0\rangle + \lambda_1 \langle 1|\psi\rangle\langle\psi|1\rangle \\ &= \lambda_0 \langle\psi|0\rangle\langle 0|\psi\rangle + \lambda_1 \langle\psi|1\rangle\langle 1|\psi\rangle \\ &= \langle\psi| \underbrace{\left[\lambda_0 |0\rangle\langle 0| + \lambda_1 |1\rangle\langle 1|\right]}_{?? = Z} |\psi\rangle = \langle\psi|Z|\psi\rangle \end{aligned}$$

Z is an example of a more general concept: an observable.

Observables

Generally, we are interested in measuring real, physical quantities. In physics, these are called observables.

Mathematically, an observable M is a Hermitian operator (matrix)

$$M = M^\dagger$$

$$UU^\dagger = \mathbb{1}$$
unitary

example: Hadamard

Analytically, the **expectation value** of measuring the observable M given the state $|\psi\rangle$ is

$$\langle M \rangle = \langle \psi | M | \psi \rangle$$

M : $|v_1\rangle$ λ_1
$|v_2\rangle$ λ_2

Why Hermitian? real eigenvalues!

Observables

Example:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

X is Hermitian and its (normalized) eigensystem is

$\lambda_0 = +1 \qquad |v_0\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

$\lambda_1 = -1 \qquad |v_1\rangle = |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

Observables

Example:

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Y is Hermitian and its (normalized) eigensystem is

$\lambda_0 = +1$ $\quad |v_0\rangle = |p\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + i|1\rangle\right)$

$\lambda_1 = -1$ $\quad |v_1\rangle = |m\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle - i|1\rangle\right)$

Expectation values: analytical

Exercise: consider the quantum state

$$|\psi\rangle = \frac{1}{2}|0\rangle - i\frac{\sqrt{3}}{2}|1\rangle.$$

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 1/2 \\ -i\sqrt{3}/2 \end{pmatrix}$$

$$Y|0\rangle = i|1\rangle$$
$$Y|1\rangle = -i|0\rangle$$

Compute the expectation value of Y:

$$\begin{aligned}
\langle Y \rangle &= \langle \psi | Y | \psi \rangle \\
&= \left(\frac{1}{2}\langle 0| + i\frac{\sqrt{3}}{2}\langle 1|\right) Y \left(\frac{1}{2}|0\rangle - i\frac{\sqrt{3}}{2}|1\rangle\right) \\
&= \left(\frac{1}{2}\langle 0| + i\frac{\sqrt{3}}{2}\langle 1|\right)\left(\frac{1}{2}i|1\rangle - \frac{\sqrt{3}}{2}|0\rangle\right) \\
&= -\frac{\sqrt{3}}{4}\langle 0|0\rangle - \frac{\sqrt{3}}{4}\langle 1|1\rangle \\
&= -\frac{\sqrt{3}}{2}
\end{aligned}$$

Expectation values

Let's do this in PennyLane instead:

```
dev = qml.device('default.qubit', wires=1)

@qml.qnode(dev)
def measure_y():
    qml.RX(2*np.pi/3, wires=0)
    return qml.expval(qml.PauliY(0))
```

Multi-qubit expectation values

Example: operator $Z \otimes Z$.

Eigenvalues are computational basis states:

$$
\begin{aligned}
(Z \otimes Z)|00\rangle &= |00\rangle \\
(Z \otimes Z)|01\rangle &= -|01\rangle \\
(Z \otimes Z)|10\rangle &= -|10\rangle \\
(Z \otimes Z)|11\rangle &= |11\rangle
\end{aligned}
$$

To compute an expectation value from data:

$$
\langle Z \otimes Z \rangle = \frac{1 \cdot n_1 + (-1) \cdot n_{-1}}{N}
$$

Multi-qubit expectation values

Example: operator $X \otimes I$.

Eigenvalues of X are the $|+\rangle$ and $|-\rangle$ states:

$$(X \otimes I)|+0\rangle = |+0\rangle$$
$$(X \otimes I)|+1\rangle = |+1\rangle$$
$$(X \otimes I)|-0\rangle = -|-0\rangle$$
$$(X \otimes I)|-1\rangle = -|-1\rangle$$

Fun fact: All Pauli operators have an equal number of $+1$ *and* -1 *eigenvalues!*

Multi-qubit expectation values

How to compute expectation value of X from data, when we can only measure in the computational basis?

Basis rotation: apply H to first qubit

$$
\begin{aligned}
(H \otimes I)(X \otimes I)|+0\rangle &= |00\rangle \\
(H \otimes I)(X \otimes I)|+1\rangle &= |01\rangle \\
(H \otimes I)(X \otimes I)|-0\rangle &= -|10\rangle \\
(H \otimes I)(X \otimes I)|-1\rangle &= -|11\rangle
\end{aligned}
$$

When we measure and obtain $|10\rangle$ or $|11\rangle$, we know those correspond to the -1 eigenstates of $X \otimes I$.

Hands-on: multi-qubit expectation values

Multi-qubit expectation values can be created using `@` :

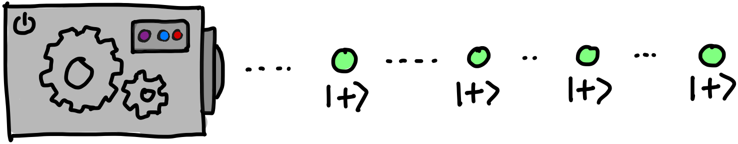
```
@qml.qnode(dev)
def circuit(x):
    qml.Hadamard(wires=0)
    qml.CRX(x, wires=[0, 1])
    return qml.expval(qml.PauliZ(0) @ qml.PauliZ(1))
```

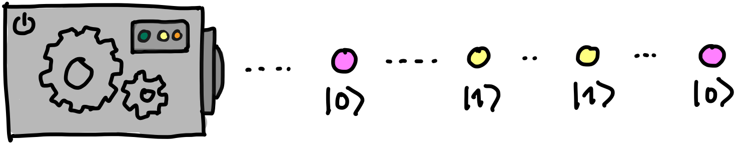
Can return *multiple* expectation values if no shared qubits.

```
@qml.qnode(dev)
def circuit(x):
    qml.Hadamard(wires=0)
    qml.CRX(x, wires=[0, 1])
    return qml.expval(qml.PauliZ(0)), qml.expval(qml.
                                    PauliZ(1))
```

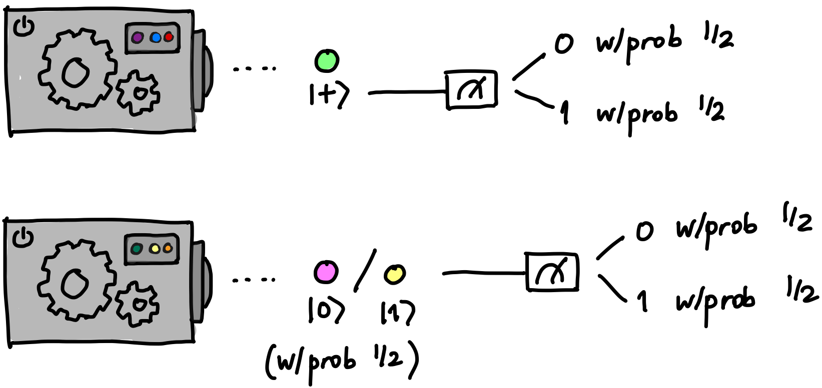
Mixed states

Are these two devices the same?

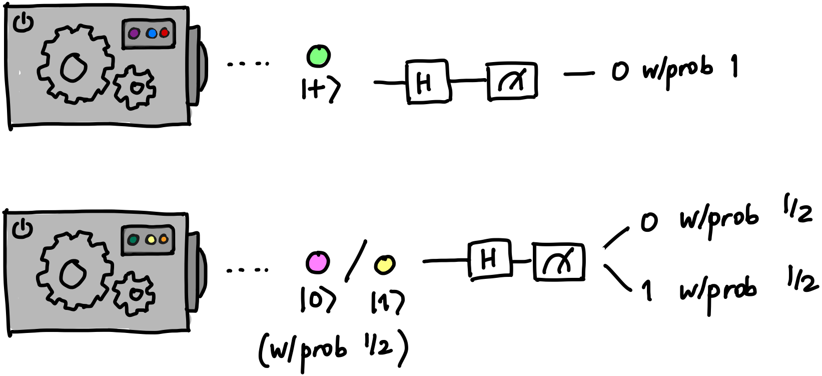




Mixed states



Mixed states



What is the second box doing?

Mixed states

The second box prepares a **mixed state**.

A **pure state** can be expressed as a single ket vector, e.g.,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

A **mixed state** is a *probabilistic mixture of pure states*. It is represented by a **density matrix**.

Density matrices

The density matrix of a pure state $|\psi\rangle$ is

$$\rho = |\psi\rangle\langle\psi|$$

Exercise: what are the density matrices for $|0\rangle$ and $|1\rangle$?

$$\rho_0 = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\rho_1 = |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Density matrices

Density matrices of mixed states are linear combinations of those of pure states:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| \qquad \sum_i p_i = 1$$

Exercise: A system prepares $|+\rangle$ with probability $1/3$, and $|0\rangle$ with probability $2/3$. What is its state?

Box 1

$$\rho_+ = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\rho_- = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

Box 2

50% $|0\rangle$

50% $|1\rangle$

$$\rho = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Density matrices

Density matrices have some nice properties.

- they are Hermitian
- they have trace 1
- they are positive semi-definite (all eigenvalues are ≥ 0)
- (for pure states only) they are projectors, i.e., $\rho^2 = \rho$

Check with our example:

Fun activity: show properties hold for general $\rho = \sum_i p_i |\psi_i\rangle \langle\psi_i|$

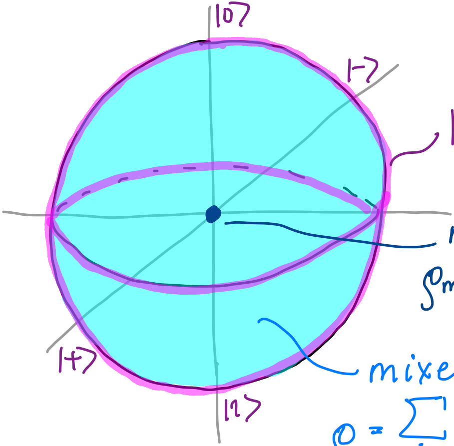
Density matrices and expectation values

Exercise: A system prepares $|+\rangle$ with probability $1/3$, and $|0\rangle$ with probability $2/3$.

$$\rho = \begin{pmatrix} \frac{5}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} \end{pmatrix}$$

What are the expectation values $\langle X \rangle$,$\langle Y \rangle$, $\langle Z \rangle$?

Density matrices and the Bloch sphere



pure states

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

maximally mixed state

$\rho_{max} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\mathbb{1}$

mixed states

$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$

Next time

Next class:

- Operations and measurements on mixed states
- Quantum channels

Action items:

1. Work on project (remember to fill out weekly surveys!)

Recommended reading:

- Codebook module NT
- Nielsen & Chuang 2.4, 2.2.5-2.2.6